

Day 03: Inference for SLR

Data for today

We're returning to the NHANES sitting time and depression risk score data. We're analyzing the relationship between depression risk score and sitting time.

i Theoretical Model:

$$\text{_____} = \beta_0 + \beta_1 x_i + \varepsilon_i \text{ where } \varepsilon_i \stackrel{iid}{\sim} N(0, \text{_____})$$

i Fitted Model:

$$\text{_____} = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

```
summary(nhanes_lm)
```

Call:

```
lm(formula = dep_score ~ sedentary_hours, data = nhanes)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.2812	-0.9671	0.0059	0.6424	2.1370

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.90426	0.07359	12.29	<2e-16 ***
sedentary_hours	0.03141	0.01155	2.72	0.0067 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

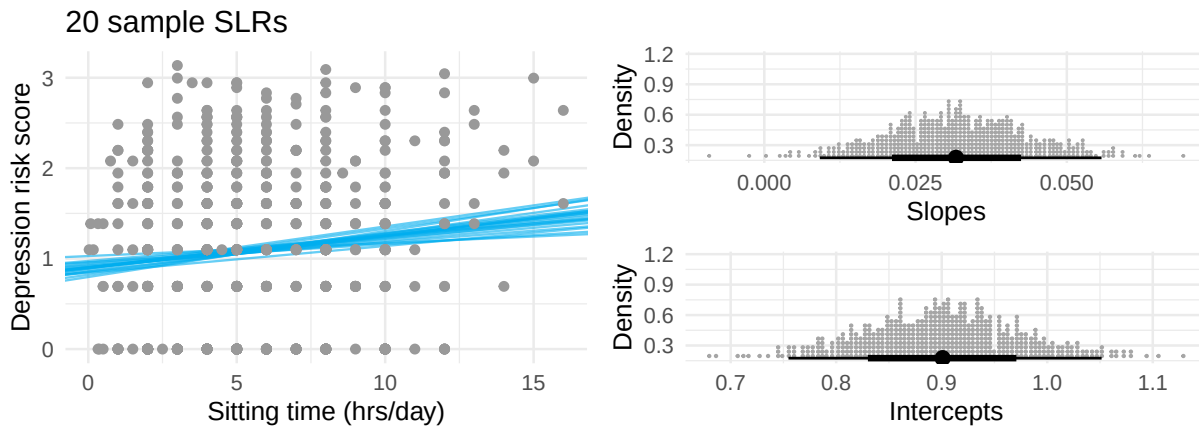
Residual standard error: 0.8835 on 657 degrees of freedom

Multiple R-squared: 0.01113, Adjusted R-squared: 0.009629

F-statistic: 7.398 on 1 and 657 DF, p-value: 0.006704

Sampling variability

Every new sample gives us a slightly different fitted line.



Sampling distribution of a coefficient

We use the **t distribution** to model the sampling distribution of a regression coefficient.

- Degrees of freedom: $df = \text{_____}$
- Mean: $= \hat{\beta}_i$
- Standard deviation: $SE(\hat{\beta}_i) = \frac{\hat{\sigma}}{\sqrt{\sum(x_i - \bar{x})^2}}$

Confidence interval

General Form:

$$\text{statistic} \pm \text{critical value} \times \text{standard error}$$

Form for regression coefficient:

$$\hat{\beta}_i \pm t^* \cdot SE(\hat{\beta}_i)$$

i Critical value

t^* is the value that cuts off the right tail area that matches your confidence level. For a 95% CI, find t^* so that the two-sided tail area is 0.025 on each side, using `qt(.025, df = n-2)`.

Hypothesis test

$$H_0 : \beta_i = \text{null value} \quad \text{vs.} \quad H_a : \beta_i \begin{cases} \neq \\ < \\ > \end{cases} \text{null value}$$

$$\text{test statistic} = \frac{\text{estimate} - \text{null value}}{SE} = \frac{\hat{\beta}_i - \text{null value}}{SE(\hat{\beta}_i)}$$

Then find the appropriate tail area(s) under the t-distribution with $df = n - 2$.

```
tidy(nhanes_lm)
```

```
# A tibble: 2 x 5
  term          estimate std.error statistic  p.value
<chr>          <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)    0.904    0.0736    12.3 2.18e-31
2 sedentary_hours 0.0314    0.0115     2.72 6.70e- 3
```

 Try it

Verify the test statistic for $H_0 : \beta_1 = 0$ against $H_A : \beta_1 \neq 0$ using the table above.

Example: 90% confidence interval for β_1

$$\hat{\beta}_i \pm t^* \cdot SE(\hat{\beta}_i)$$

$t^* =$

```
qt(0.95, df = 659 - 2)
```

```
[1] 1.647176
```

$$\hat{\beta}_i \pm t^* \cdot SE(\hat{\beta}_i) = 0.031 \pm 1.647 \cdot NA \implies (0.012, 0.05)$$

Interpretation: We are 90% confident that for every 1-hour increase in sitting time, the depression risk score changes by between 0.012 and 0.05, on average.

💡 General template (reusable for any CI)

- We are _____% confident that for every 1 _____ increase in _____,
- _____ also _____
- by between _____ and _____

Discussion: Recall from Day01: we found no significant difference between the “High” and “Low” sitting groups. How does this result compare? Which do you trust more?